

SQL Set Operations: UNION, INTERSECT, and EXCEPT

Introduction

SQL Set Operations are used to combine or compare the results of two or more SELECT statements. The three main set operations are:

- UNION
- INTERSECT
- EXCEPT (called MINUS in Oracle)

These operations are commonly used in reporting, data analysis, and business intelligence applications.

1. UNION

Definition

The UNION operator combines the results of two SELECT statements into a single result set and removes duplicate rows automatically.

When to Use

- When combining data from multiple tables.
 - When creating consolidated reports.
 - When duplicate records should appear only once.
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Syntax

```
SELECT column1, column2  
FROM table1
```

```
UNION
```

```
SELECT column1, column2  
FROM table2;
```

Rules

1. Both SELECT statements must return the same number of columns.
 2. Corresponding columns must have compatible data types.
 3. Column names in the final result are taken from the first SELECT statement.
 4. Duplicate rows are removed automatically.
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Example

```
SELECT department_id  
FROM employees
```

```
UNION
```

```
SELECT department_id  
FROM departments;
```

Result

Returns a list of unique department IDs from both tables.

2. INTERSECT

Definition

The INTERSECT operator returns only the rows that exist in both result sets.

When to Use

- To find common records between tables.
 - To identify overlapping data.
 - To compare datasets.
-

Syntax

```
SELECT column1, column2  
FROM table1
```

```
INTERSECT
```

```
SELECT column1, column2  
FROM table2;
```

Rules

1. Both queries must return the same number of columns.
 2. Data types must be compatible.
 3. Only matching rows from both result sets are returned.
 4. Duplicate rows are removed automatically.
-

Example

```
SELECT department_id  
FROM employees
```

```
INTERSECT
```

```
SELECT department_id  
FROM departments;
```

Result

Returns department IDs that exist in both EMPLOYEES and DEPARTMENTS tables.

3. EXCEPT (Oracle: MINUS)

Definition

The EXCEPT operator returns rows from the first query that do not exist in the second query.

Note: Oracle uses the keyword MINUS instead of EXCEPT.

When to Use

- To identify missing records.
 - To compare differences between datasets.
 - To find unmatched values.
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Syntax (Standard SQL)

```
SELECT column1, column2  
FROM table1
```

```
EXCEPT
```

```
SELECT column1, column2  
FROM table2;
```

Oracle Syntax

```
SELECT column1, column2  
FROM table1
```

```
MINUS
```

```
SELECT column1, column2  
FROM table2;
```

Rules

1. Both SELECT statements must return the same number of columns.
 2. Data types must be compatible.
 3. Only rows from the first query that are not found in the second query are returned.
 4. Duplicate rows are removed automatically.
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Example

```
SELECT department_id  
FROM departments
```

MINUS

```
SELECT department_id
FROM employees;
```

Result

Returns departments that currently have no employees assigned.

Comparison of Set Operations

Operation	Purpose	Removes Duplicates
UNION	Combines results from both queries	Yes
INTERSECT	Returns common rows	Yes
EXCEPT / MINUS	Returns rows in first query but not second	Yes

Conclusion

SQL set operations provide powerful methods for combining and comparing result sets. UNION is used to merge data, INTERSECT identifies common records, and EXCEPT (MINUS in Oracle) finds differences between datasets. Understanding these operations is essential for building business reports, performing data analysis, and supporting enterprise database solutions.